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COURSE: AGT 102- Annual Crops

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A crop is any useful plant, or a plant which is grown for any purpose. It is utilized by people directly or indirectly, raw or processed. These plants are intentionally grown or managed for various uses.

Annual crops are plants that complete their life cycle, from seed germination to seed production, within a year or a single growing season. Annual plants germinate, form a seed, grow to maturity, flower, produce fruit and seed, and then die all within a single year. They are usually characterised by bright flowers to attract pollinators within their short life cycle.

Most topical arable crops can be grouped into ten main families, namely:

Agavaceae	-	Isal
Araceae	-	Cocoyam
Convovulaceae	-	Sweet potato
Dioscoreaceae	-	Yam
Euphorbiaceae	-	Cassava
Fabaceae	-	Cowpea, groundnut, soybean and pigeon pea
Malvaceae	-	Cotton, okra and kenaf
Poaceae	-	Maize, rice, millet, sorghum and wheat
Solanaceae	-	Irish potato and tobacco
Tiliaceae	-	Jute

Classification of Agricultural crops

The International Code of Botanical Nomenclature (ICBN) explains the principal ranks of taxa. According to them, plants are classified based on the following criteria: (1) botanical, (2) descriptive, and (3) agricultural. Globally, major crop classifications are defined by the Food and Agriculture Organization (FAO) of the United Nations. For a better understanding of agricultural crops and their distribution in Nigeria and the globe, knowledge of the classification of plants and crops is necessary. Crops can be classified based on their

1. Morphology 2. Culture 3. Growth period 4. Mode of production 5. Uses

A. Agricultural crops can be classified into two based on their Morphology.

I. Monocotyledonous Crops: These are plants that have only one seed leaf with narrow leaf variation. Examples include maize, millet, guinea corn and wheat.

II Dicotyledonous Crops: These are plants that have two seed leaves with broad leaves. Examples are groundnut, cowpea, soyabean, pigeon pea etc.

B. There are two classes of crops based on culture.

I. Cultivated Plants: They are also known as domesticated plants. They are plants or plant species that have arisen through human action like selection or breeding, and they depend on management for their continued existence.

II. Wild Plants: Plants that grow spontaneously in self-maintaining populations in natural or semi-natural ecosystems and can exist independently of direct human action.

D. Crops can be classified into three based on their life span or duration of growth. Life span or life cycle is the length of time from planting of a seed through its germination, maturity, flowering, and seed production to its death.

I. Annual Crops: Annuals crops are plants which live within a short period of time, for a few weeks or months, perpetuated by seed, and which die soon after producing seeds. Examples include melon, cotton, cowpea, maize, rice, wheat, groundnut etc.

II. Biennial Crops: Biennial crops are plants that require two years or two growing seasons to complete their life cycle, the first for vegetative growth and accumulation of food reserves, and the second for the production of reproductive parts. Examples include: onion, cabbage, carrot, celery, parsley and leek.

III. Perennial Crops: Perennial crops are plants that grow and mature in more than two years, including all trees and shrubs and many herbaceous plants with underground stems (e.g. corm, rhizome, tuber) like banana and clump-forming grasses. The Perennial plants continue growing and produce seeds year after year, either from a single plant or, in herbaceous plants, from succeeding regrowth. There are two types of perennial crops:

[1]. Woody perennials that live for 3-10 years e.g paw-paw

[2]. Herbaceous perennials that live for 30-80 years e.g cocoa, oil palm, mango, orange, rubber etc.

E. Based on the mode of production, crops can be classified into four.

I. Field cash crops: The Agronomic crops are also called as “field crops”. They are mostly annual herbaceous plants that are grown under extensive or large-scale culture.

II. Forage crops: The Pasture and Forage Crops are very essential varieties of plants. These plants are grown or managed as vegetable feed for grazing animals.

III. Horticultural crops: Horticultural crops have been referred to as “garden crops”. Horticultural crops are annual and perennial plants which are grown under an intensive system of

agriculture. The following are the major kind of horticultural crops: Olericultural or Vegetable Crops, Pomological or Fruit Crops and Nuts, Spice Crops, Essential-oil Crops, Ornamental Crops, Biocidal Crops, Industrial Crops, Plantation Crops.

IV: Plantation crops: There are two more categories of plantation crops. One is Biocidal Crops and the other is Industrial Crops. The Biocidal Crops are plants containing organic compounds with pesticidal or anti-microbial properties. The effective parts are either directly applied or seeped in water for foliar spray. The Industrial Crops are plants grown to provide materials for industrial processing and production of non-food products, including drugs.

The crops used in the farmland are also included under the plantation crops category. They include: 1) Main Crop 2) Nursery crop 3) Intercrop 4) Filler Crop 5) Relay Crop 6) Companion crop 7) Green manure crop 8) Trap crop or Decoy crop 9) Insect pest repellent crops 10) Natural enemies attractant crops.

- Main Crop is any crop which is intended by the grower to become his or her main source of revenue.
- Nursery crop is a plant that is temporarily grown in the nursery and later planted in the field or garden or used for ornamental display, when it reaches the proper age and size.
- The Intercrop is any crop which is planted simultaneously with or before the flowering season of the main crop in intercropping.
- Filler Crop is any crop which is planted to fill a gap.
- Relay Crop is the crop which is planted after the flowering period or harvest of the main crop in relay cropping.
- Companion crop is any crop which is planted close to the main crop to complement the latter's growth and production, or to maximize utilization of space because they do not compete.
- Cover crop is a crop grown mainly to control soil erosion, regulate soil temperature, control weeds and reduce evaporative losses.
- Green manure crop is a leguminous crop grown to be plowed under the soil to increase organic matter and serve as organic fertilizer. Mungbean (mungo) is ideal for this purpose.
- Trap crop or Decoy crop are plants grown to attract certain insect pests or parasites because they are favorite hosts. They act as decoys to lure pests away from the main crop.
- Insect pest repellent crops are plants grown along the borders and at strategic places in the farm to repel insect pests because of their strong aroma and anti-herbivory properties.
- Natural enemies attractant crops are flowering plants grown at strategic places in the farm to attract natural enemies of insect pests.

F. The agricultural classification of plants/ crops based on their primary uses include the following classes: 1. Food Crops 2. Cash Crops 3. Staple Crops 4. Cereal or Grain Crops 5. Legume Seed Crops or Pulses 6. Root and Tuber Crops or Tuberous Crops 7. Oil Seed Crops 8. Sugar and Sweetener Crops 9. Beverage Crops 10. Rubber Crops 11. Latex and Gum Crops 12. Dye and Tannin Crops 13. Fiber Crops 14. Pasture and Forage Crops 15. Biofuel Crops 16. Vegetable

- The Food Crops are plants grown primarily for the harvesting of any part which is used by people as food or processed into food product.
- The Non-food Crops are mainly cash crops. These are plants grown for the production of non-food products such as fiber, fodder, alcohol and tobacco.
- The Staple Crops are food crops. These are regularly consumed in a traditional diet and from which people obtain a major proportion of their energy and nutrient requirements. Examples of other staple crops: wheat and millet.
- The Cereal or Grain Crops are mainly annual, herbaceous plants belonging to the grass family which are grown for their seeds or grains. Examples: corn, millet, rice, sorghum, wheat.
- The Legume Seed Crops or Pulses are leguminous plants which produce edible, protein-rich seeds. Examples: cowpea, peanut, soyabeans and peas.
- The Root and Tuber Crops are plants with modified, swollen root or underground stem. These organs are rich sources of carbohydrate and are commonly used as staple, livestock feed, or as raw materials for industrial purposes, such as starch and alcohol production, or processed into various food products. They include cassava, yam, cocoyam, sweet potatoes and irish potato.
- The Oil Seed Crops are plants grown for their seeds which are rich source of edible and industrial oil. Example: sunflower
- The Sugar and sweetener Crops are plants grown primarily for the production of sugar or other sweet- tasting products. Various forms of sugar are derived from various parts of certain plants such as from stems, bulb and other underground organs, leaves, flowers, fruits, seeds, sap and resin. These are called as sugar crops.
- The Beverage Crops are plants which are sources of various drinks including fruit juices. Examples: cocoa, coffee and tea.
- The Rubber Crops are another major type of plants that are grown for the production of latex which is processed into the industrial product called rubber.
- The Latex and Gum Crops are very typical plants. These plants are grown for the collection of latex which is processed into chewing gum, inelastic rubber and other industrial products such as surgical tapes and dental supplies, insulation, splints, pipes, golf balls, waterproofing, adhesives, etc.
- The Dye and Tannin Crops are special kinds of plants. These plants are grown as sources of tannin and coloring substances. Tannin is an aromatic, phenolic substance which is obtained from barks and other plant organs and variously used in tanning, medicines, dyeing, ink manufacture, etc.
- Fiber Crops are plants grown as sources of fiber used in making textiles, ropes, twine and similar materials. The fiber is extracted from the bark, leaves, or other organs including the husk of coconut. Examples are cotton, sisal, kenaf, hibiscus etc
- The Pasture and Forage Crops are very essential varieties of plants. These plants are grown or managed as vegetable feed for grazing animals.
- Soilage Crops are the grasses grown, cut and directly fed to animals. Silage Crops are the grasses grown, cut, fermented and preserved before being fed to animals.
- Spices: They are those crops that give flavour to our food e.g onions, ginger, garlic, pepper etc.

- The Biofuel Crops are special type of crops. These plants are grown for the production of fuel that is used as additive or replacement for petroleum products. Examples: sugarcane, corn, coconut, castor bean and *Jatropha*.
- Vegetable Crops are (except mushroom) grown for their succulent and edible parts such as the roots, stems, leaves, young tops, fruits or seeds for use in culinary preparations either fresh or preserved in the fresh state.

Vegetables are classified based on their edible parts, into several varieties. They are:

- a. Leafy Vegetables are crops that are grown mainly for their leaves. Examples: amaranth, lettuce, and raddish.
- b. Shoot Vegetables are plants grown primarily for their edible shoot, mainly the young, succulent stem. Examples: asparagus and celery.
- c. Pod and Seed Vegetables are generally members of *Leguminosae* or *Fabaceae* family. These plants are grown for their young pods and seeds. Examples: beans and sweet corn.
- d. Root and Bulb Vegetables are plants grown for their swollen underground roots and stems. Examples: carrot, potato, onion and raddish.
- e. Flower Vegetables are plants with edible flowers. Examples: rose and sunflower.
- f. Fruit Vegetables are grown for their fleshy, succulent fruits. Examples: eggplant, tomato, peppers and melons.

Vegetables are also classified according to their families, into the following groups:

- a. Cole Crops or Crucifers. These vegetable crops belong to the *Cruciferae* or *Brassicaceae* (Mustard) family. They possess edible leaves or heads. Examples: cabbage, cauliflower, mustard, and raddish.
- b. Cucurbits, belong to the Gourd and Squash family. They are grown mainly for their fruits but some have edible young shoots and flowers; also called Vine Crops. Examples: bottle gourd, cucumber and melons.
- c. Legume Vegetables are the members of the *Leguminosae* or *Fabaceae* (Bean) family. The seeds are rich in protein. Examples: kidney beans, pea and pole sitao.
- d. Lilies are members of the *Liliaceae* (Lily) family. Examples: asparagus, garlic and onion.
- e. Solanaceous Crops, belong to the *Solanaceae* (Nightshade or Eggplant) family and, with the exception of white potato, are also called Fruit Vegetables. Examples: eggplant, tomato, peppers and white potato.

f. Mushrooms are edible fungi belonging to the division *Basidiomycota* (club fungi). The edible part commonly consists of an upright stalk and an umbrella- shaped cap.

CEREAL CROPS (Crops of the Grass Family)

MAIZE (*Zea mays* L.) **2n = 20**

Family: Poaceae

Origin: Maize originated in the new world, around Mexico or Central America.

Ecology: It is grown across a wide range of latitudes and from sea level to over 3000m in the Andes and Mexico. It is mostly grown in regions with a temperature range of 21-30⁰c. It does well in sandy loam soils, rich in nutrients, especially nitrogen.

Botany: Maize is a stout annual grass and like most plants in the grass family, the roots are adventitious and arise from the lowest nodes of the stem below ground level. Maize usually has a single stem and unlike most grasses, the stalk (culm) is solid. The leaves are sessile and are made up of a leaf blade, sheath and collar-like ligule. The maize plant is monoecious with male and female inflorescences borne in separate positions on the same plant. The male or staminate inflorescence is a panicle called 'tassel' which terminates the main axis. The female inflorescence is known as the 'ear' and is carried in the axil of the maize leaf just below the middle part of the plant. Maize is normally cross-pollinated; it is protandrous with pollen being shed before the silks are receptive, but there is some overlap and up to 5% self pollination can occur. The fertilized ovary develops into a caryopsis called 'kernel' or 'grain'. The so called maize seed is really a fruit as the pericarp and testa, which are both thin, are fused together to form a single membrane. There are seven types of maize-pod, pop, flint, dent, flour, sweet and waxy. Most maize cultivars belong to the dent and flint groups; the dent cultivars are those with the greatest yield potential.

Varieties: Some of the well known improved varieties in Nigeria are the 'TZ series' developed at the International Institute for Tropical Agriculture (IITA) and the 'FARZ series' developed at the former Federal Department of Agricultural Research (FDAR) now National Cereals Research Institute. Examples are SWAN-1, OBASUPER-1, OBASUPER-12, TZSR – W, TZSR – Y, TZESR – Y, TZESR-W, FARZ 27, FARZ 34 and several hybrids.

Cultural Practices: The seed rate is 25-35 kg/ha and the recommended spacing is 75 x 25cm, which will give a plant population of 53,333 plants/ha.

Planting: Planting is done in March/April for the first (early) crop and in August for the second (late) crop.

Weed control: With regard to weed control, atrazine based herbicides are recommended at the rate of 3kg a.i./ha. This should be applied pre-emergence, at planting or 1 or 2 days before or

after planting. In the absence of herbicide, two weedings at 3 and 7 weeks after planting should suffice.

Fertilizer application: The recommended fertilizer is NPK (15-15-15) at the rate of 300kg/ha, preferably applied at 2-3 weeks after planting. Nitrogen can also be applied as ammonium sulphate, urea and calcium ammonium nitrate at the rate of 100 kg N/ha. Phosphorous can be applied as single or triple super phosphate at the rate of 50kg P₂O₅/ha and potassium as potassium chloride or sulphate at the rate of 50kg K₂O/ha.

Harvesting: A green harvest can be done when the crop is physiologically mature. A dry harvest can then be taken by dehusking or with the husks still attached. Harvesting can be done mechanically with cutlass or by hand.

Yield: The yields from farmers' plots are low, about 200-2000kg/ha of dry grains. Yields of up to 5000 kg (5t)/ha or more of grain are possible using improved practices.

Uses: Uses of maize can be grouped into three: (i). staple human food. (ii). feed for livestock and (iii). Industrial raw material

Diseases of maize

- (1) **Maize rusts**
- (2) **Maize leaf blight**
- (3) **Curvularia leaf spot**
- (4) **Brown spot**
- (5) **Downy mildew**
- (6) **Maize Streak Virus (MSV)**

Insect Pests: Insect pests of maize include stem borers-*Busseola fusca* and *Sesamia calamistis*. Storage pests which affect maize are *Sitophilus zeamais*, *Sitotroga cerealella* and *Tribolium castaneum*. *Sitophilus zeamais*.

LEGUMES

COWPEA {*Vigna unguiculata* (L.) Walp} 2n = 22, 24

Family: Fabaceae

Origin: It probably originated in Central Africa.

Ecology: They tolerate heat and relatively dry conditions. They can be grown on very poor acid soils as a soil improver. They are photoperiod sensitive and are classified as short day plants

Botany: It has a short taproot with numerous spreading laterals, which possess nodules. The nodules are capable of fixing atmospheric nitrogen; up to 30-70 kg/ha/year. The stem is viny or semi viny with many trifoliate leaves. The leaflets are relatively smooth and shiny. The white or

purple flowers are borne in pairs in short racemes. The plants are largely self-pollinated but some cross-pollination occurs. The growth habit of cowpea is indeterminate. The pods are smooth and vary in length depending on variety. The seeds are variable in size, shape and colour.

Varieties: Some of these are Vita 5 and IT84E – 108 which are white; Ife brown, IT84E 124 and IT84S 2246 - 4 which are brown. The latter (IT84S 2246 - 4) is resistant to aphids, thrips, bruchids, etc.

Cultural practices: The first or early crop is sown when the rains are reliable in April and the second or late crop is sown at the end of August. In the North, where only one crop is possible, planting is done in June or early July. Planting of seeds should be at a spacing of 60 x 25cm or 60 x 30cm on the flat at a seedrate of 30-45kg/ha. Weeds can be controlled manually at 3 and 7 weeks after planting. Herbicides can also be used; e.g. galex at 2-3 kg a.i./ha, applied pre-emergence or imazaquin at 0.2-0.25kg a.i./ha applied pre-emergence or at early post emergence within 14 days after planting. The crop should be sprayed with nuvacron 60 a 30ml/9 litres of water weekly from the period of flowering. The plants are in any case able to fix atmospheric nitrogen. However, in every poor soils, a starter dose of 20kg/ha of N may be applied. Cowpea responds to phosphorus and 20-40kg/ha P₂O₅ as single superphosphate could be applied. General fertilizer application may also be done with 200kg/ha of NPK (15.15.15) just before planting.

Harvesting: The pods are hand-picked and because the pods ripen unevenly, 4-5 pickings may be necessary.

Yield: Under local conditions, cowpea yields vary widely between 250 and 1000kg/ha of dry grains. With good management and adopting recommended practices, yields of 1,500-2,000kg/ha are possible. The potential grain yield is estimated to be up to 3,500kg/ha.

Storage: The seeds can be stored in bags or sealed containers or in polythene/plastic bags. The equivalent of one tablet of phostoxin to 1000kg of grains will keep storage pests at bay.

Diseases: Diseases that affect cowpea include wilt caused by *Fusarium oxysporum*, scab by *Elsinoe phaseoli*, stem rot by *Sclerotium rolfsii*, leaf spot by *Cercospora canescens*, brown blotch by *Colletotrichum spp.*, etc. Most of these can be controlled with fungicides like Dithane M-45 (mancozeb) and Benlate (benomyl) and the use of resistant varieties.

Pests:

1. The pre-flowering pests include leaf hoppers (*Empoasca spp.*), cowpea aphid (*Aphis craccivora*), foliage beetle (*Ootheca mutabilis*) and striped foliage beetle (*Medythia quaterna*).
2. Post-flowering pests include flower thrips (*Megalurothrips sjostedti*), seed moth (*Cydia ptychora*), blister beetles (*Mylabris spp.*), pod sucking bug (*Nezara viridulua*).
3. The common storage pest is the cowpea weevil, *Callosobruchus maculatus*.

Nematodes: Rot knot nematodes (*Meloidogyne incognita*, *M. arenaria*) are widespread through the tropics.

FIBRE CROPS

COTTON (*Gossypium spp.*) $2n = 26$

Family: Malvaceae

Origin: Origin of the species of *Gossypium* has been a subject of considerable controversy. There are probably two general centres of origin of the cotton plant: Indo-China and tropical Africa in the old world, and Central and South America in the new world. Some authorities claim that the two cultivated species are of Central and South America origin.

Ecology: Cotton production extends from 37°N to 32°S . Cotton can be grown in a variety of soils from light sandy soils to heavy soils.

Botany: It has a tap root system, leaves occur spirally on the stem, the flowers are borne on specialized braches, which occur on the upper part of the plant. The fruits are spherical or ovoid leathery capsules, called bolls, 4-6cm long. The bolls of the two commonly grown species can be distinguished from each other. The seeds are dark in a colour.

Varieties: Recommended varieties include ‘Samaru 71’ of the Northern States of Nigeria and ‘Allen 26J’ for Edo North.

Cultural practices: Seeds are planted at 4-6 per hole to 25cm apart on 90cm or 75cm ridges. Most of the cotton is planted after mid-July. Weeds can be controlled manually or with herbicides; e.g. 0.8 kg a.i./ha of diluron + 0.8kg a.i./ha of norflurazone applied pre-emergence. Fertilizer needs of the crop can be met by applying recommended rates of 50kg N, 25kg P_2O_5 and 20kg K_2O /ha. Compound NPK (15.15.15) fertilizer can also be used at the rate of 200-300kg/ha.

Harvesting: Cotton is picked when dry from fully opened bolls and picking should be done regularly so that open cotton is not left in the field which might result in deterioration in colour and quality of the lint.

Yield: An average yield of 300 and 500kg/ha of seed cotton are obtained for July (late) and April (early) sown cotton, respectively.

Processing: Seed cotton from the field is first dried. It is then put into a machine, which separates the lint from the seed and fuzz. This type of machine is called a **gin**.

Uses

Cotton gives three important products which are used as fibre, food and feed material.

1. Cotton lint is used for clothing, household and industrial products. Perhaps the outstanding quality of cotton is its light tensile strength, i.e ability to withstand tearing or breaking when it is under stress. Unlike other fibres, the tensile strength is appreciably increased when it is wet.

2. Cotton seed oil, derived from the seeds, is one of the most important of the world's non-drying oils. Refined cotton seed oil is used mainly for edible purposes – for cooking, salad, manufacture of margarine, and to a lesser extent in the packing of fish and cured meat.
3. Cotton-seed cake, obtained after extraction of oil, contains 21% protein and is used for livestock feed. Cotton seed hulls are used as roughage for livestock and as livestock bedding, fertilizer, fuel and to make paper.

Diseases: Major diseases of cotton are bacterial blight, fusarium wilt and leaf curl.

Insect pests: cotton stainer (*Dysdercus supersticiosus*), Bollworms (*Heliothis armigera*, *Earias spp.*)

ROOTS AND TUBERS CROPS

YAM (Botanical name - *Dioscorea spp*)

Family name - Dioscoreaceae

Origin: The species of yam are of different origin. Most of them are West Africa and Asian Origin.

Ecology

- Temperature required is 25 – 30⁰c
- The crop requires a well distributed rainfall of at least 15mm/Year.
- Tuber formation is promoted by a short day condition.
- Yam required loosed, deep well drained loamy soils of reasonable fertility.
- In Nigeria they are grown in the South and middle belt.

Botany: The yam tuber is neither stems nor root tubers as it develops from the hypocotyl region (The transition zone between the stem and the roots). The stem of yam is a vine which can grow up to 12ft in length and is generally provided with a support for climbing. The long petiole leaves may be opposite or alternate. The flowers are small often dioecious or unisexual.

The Species of Yam

1. *Dioscorea rotundata* (white)
2. *Dioscorea cayenensis* (yellow yam)
3. *Dioscorea alata* (water yam)
4. *Dioscorea bulbifera* (aerial yam)
5. *Dioscorea esculenta* (Chinese/ Lesser yam)
6. *Dioscorea dumetorum* (Bitter/ Trifoliate yam)

Cultural Practice: In southern Nigeria yam can be planted in November while the soil is fairly moist (Early yam) and in March- April (late yam). In the north, planting is done in May-June. The seedling rate required is about 2-3t/ha of seed yam. Small tubers may be planted whole but large ones may be cut into small pieces (setts) for economy.

Weed Control: Weed control can be done manually and or with herbicide like premixtra at the rated 3kg ai/ha which should be applied pre-emergence.

Fertilizer application: Fertilizer should be applied at 6-8WAP and NPK at the rate of 300 400kg/ha in a band.

Harvesting: Yam matures in 8-12 months depending on varieties. There are two kinds of harvesting practice. These are:

1. **Double Harvesting:** In the double harvesting system, the yam plant is harvested twice. The first harvesting is done at about 5-6 months during which the tuber is carefully unearthed without damaging the roots. A second harvest is taken after the plant has dried at the end of the season.
2. **Single Harvesting:** This involves waiting till the end of the season when the vines had died back before yams are harvested.

Yield: Under farmers condition yam varies between 7-12t/ha. With good management practices, yield of up to 25t/ha is possible.

Uses

1. Mainly for human Consumption: The most common method is by boiling, pounding, roasting, frying, bakery etc.
2. Wild species have been used as a source material for the manufacture of oral contraceptives.

Disease: The main disease affecting yams are.

- a. Tuber rots caused by fungi *Penicillium spp* and *Fusarium spp*
- b. Anthracnose is a fungal disease that causes leaf spots, blight and tuber rot.
- c. Yam Mosaic Virus causes yield loss due to reduced tuber sizes.
- d. **Nematodes:** The yam nematode *Scutellonema bradys* and the root-knot nematode (*Meloidogyne spp*) also attack yam. These can be controlled by effective crop rotation which includes non-susceptible crops.

Pest: The most serious insect pest of yam is Yam beetle (*Heteroligus meles*).

SUGAR CANE (*Saccharum officinarum*)

Origin: It has its centre of variation in North guinea savanna and Indian. Sugar-cane was cultivated by the Indians for domestic use as a source of sweetening and chewing. Sugar-cane also claimed to have originated from Asia and probably due to hybridization, among its wide (varieties) relatives. *S. Spontaneum* found in India, Philippines, Malaysia, Indonesia. *S. robustum* is found mainly in New Guinea zones. Although, all the other part of the world of it centre of

origin have not be assembled. Sugar-cane was introduced to the America and Carribean Island during the East to West slave trade.

Uses of Sugar-Cane: Sugar-cane is chewed, swallowing of sugary juice which is a source of energy (Sucrose). Herbalist boils pieces of cane with medicinal preparation primarily for sweetening purpose

Type and varieties

Sugar varieties fall broadly into two (2) groups:

- A. Varieties with fertile pollen and eggs:- These are self-pollinators
- B. Varieties with sterile pollen grains but with fertile eggs:- These are cross pollinators.

There are two main type of sugar-cane:

- 1. Noble
- 2. Thin cane

Ecology: Sugar-cane is cultivated under difficult ecological zone of the tropics and sub-tropics.

Soil-requirement: Sugar-cane requires soil with high content of iron, clayey soil which is formed from seed bed silt.

Climate requirement: Cane requires idea temperature range $25^{\circ}\text{C} - 38^{\circ}\text{C}$. Rain-fall per annum should be between 1000mm and 1520mm. It requires plenty of sunlight for good growth.

Cultural Practices: Standard spacing for sugar-cane vary from 1m x 1.5 m to 1.5 m – 1.75 m depending of the vegetative characteristics of the varieties. Propagation is done by vegetative or by seed. Commercially, sugar-cane is vegetatively propagated.

Maintenance: For cropping system:- sugar- cane may be involve in crop rotation system. After the initial crop on 2 or 3 ratoon, alternate crops such as soya beans, - cowpea, and groundnut can be planted in the field, to prevent past and disease attack.

Weeding: Manual weeding with cutlass, hoe, mechanical weeding with the use of mechanical weeders, mowers, harrows etc. are necessary.

Fertilizer/manure application: The most widely use method of fertilizer application in broadcasting. The application should be done at 6 weeks and 12 weeks during earthling-up.

Harvesting: The 3 harvest are

- (A) **First (main) harvest:** Harvesting for planting is carried-out when cane is 12-15 months old.

- (B) **First ratoon crop:** After the main harvest, the stalk are allowed to grow. The regrowth produce a second crop. This practice in referred to as ratoon.
- (C) **Second Ratoon Crop:** After the 1st ratoon crop, the stumps are once again maintained for another period of 20-24months when they are harvested at second ratoon crop. After 2nd ratoon crop, the land is ploughed up and allowed to remain fallows for 20-24 months before a new planting is carried out.

Processing

Individual processing of sugar-cane is known as milling. Millable stalk are crushed in the milling machine the stages in processing are.

1. Crushing the stock (milling)
2. Extraction of the Juice:- This process separate the juice 4m of bargasse (shelf)
3. Concentration of juice through centrifugation: this separates the crystalline sugar 4m the molasses.
4. Drying of the crystalline sugar which could be pack for sale either as grinded brown sugar or cube brown sugar.
5. Belachign followed by packing for sale as ground or cube sugar.

VEGETABLES

Vegetables apply to plant parts usually eating fresh or cooked with other food in course of a meal or used for decorations.

Amaranthus (Botanical name: *Amaranthus spp*)

Family: Amaranthaceae

Origin: The crop is known to originate from South America or Mexico.

Ecology: Amaranthus can grow well in soils with high organic matter with pH of 5.5-7.5. It requires high level of nitrogen. The temperature range for good growth is 27-30°C.

Botany: Amaranthus is a short leaved annual plant that can grow to a height of 1m. The stem is thick and fleshy with lanceolate leaves which may be green or purple in colour depending on the cultivar. The flowers are small and white to yellow in colour. The seeds are also very small. Some examples of amaranthus are:

- a. *Amaranthus cruentus*
- b. *A. caudatus*
- c. *A. dubius*

Cultural practices: The seeds are usually sun broadcast and could be mixed with soil because they are very small. The area can be mulched with dried grasses, which should be removed when the seedlings emerged. The seedlings are then transplanted to the field at about 10-12 days.

Planting in the field is done at the spacing of 30 x 30cm. It is advisable to apply organic manure at the rate of 5-10t/ha or fertilized with NPK, at the rate of 200kg/ha.

Weed control: Weed control is done manually at 3 and 6 weeks after transplanting

Harvesting: The harvesting of amaranthus can be carried out at 30-50 days from planting depending on the variety when the plant is about 15-20 cm tall. Harvesting is done by uprooting the whole plant and cutting back to 15 cm if you required subsequent harvest.

Yield: The yield of amaranthus is around 20-25t/ha of leaves and shoots weight. Seed yield is about 200kg/ha because they are small.

Uses: it is mainly for cooking food.

Disease: The diseases are mainly foliar and may include:

1. Leaf spot caused by *Cercospora spp* or *Cladosprium spp*
2. Damping off caused by *Pythium spp*

Field handling and processing of harvested annual crops

Getting crops planted and harvested at the right time are important for fruit and vegetable production, but the key to a successful operation is how the crop is handled immediately after harvest.

Post-harvest handling is the stage of crop production immediately after harvest, including cooling, cleaning, sorting and packing the crop for later stages in the process such as storage, transportation and delivery to the customer. These post-harvest handling practices are termed primary processing and they include cleaning, grading, decorticating, milling, pounding, grinding, tempering, parboiling, soaking, drying and sieving. How a product is handled during this crucial period largely determines its quality and safety for consumption.

A farmer's primary concerns during post-harvest handling are to quickly reduce the field temperature of the crop to extend shelf life and retain nutrient value; minimize moisture loss and shrinkage; and avoid physical damage such as bruising to delay spoilage. Sanitation is essential in all steps of the process. The post-harvest system can be split into three distinct areas. The first is the preparation of harvested crops for storage. The second, which is referred to as primary processing, involves further treatment of the crop to clean it, remove the husk or reduce the size. The products from primary processing are still not consumable. The third stage (secondary processing) transforms the crops into edible products. Primary processing of cereals includes cleaning, grading, hulling, milling, pounding, grinding, tempering, parboiling, soaking, drying, sieving.

Postharvest treatment for storage/ primary processing of crops

1. Destalking: Destalking or destemming is the removal of the stalk or stem from fresh fruits or vegetables.
3. Threshing: Threshing is the process in which the edible grains are separated from the straw to which they are attached either by trampling on the grain, beating it with sticks or by the use of threshing machine. It is the step in grain preparation after reaping. It involves three different operations: Separating the grain from the panicle; sorting the grain from the straw; winnowing the chaff from the grain. Separation of the grain from the panicle is the most energy-demanding of the three processes. It is the first process to have been mechanised. Sorting the grain from the straw is relatively easy, but is difficult to mechanise. Winnowing is relatively easy, both by hand and by machine. Today, threshing is mostly done by a machine such as a combine harvester which harvests, threshes and winnows the grain while it is still in the field.
4. Winnowing: Winnowing is the process of separating the outer cover or chaff or straw from the grain. This process helps in the removal of weevil or other pest in the grain. It can be done manually by dropping the grain and chaff from a height and letting the wind blow away the lighter chaff while the heavier grains fall backdown for recovery. Techniques included using Hand-held winnowing baskets are used to shake the seeds to separate out the dirt and chaff. They are very effective, but slow. Fanning mills or a winnowing fan (a shaped basket shaken to raise the chaff) or using a tool (a winnowing fork or shovel) on a pile of harvested grain.
5. Sorting: Sorting is the separation of crops into categories on the basis of a measurable physical property. Like cleaning, sorting should be employed as early as possible to ensure a uniform product for subsequent processing. Sorting also allows the separation at first sight of some undesirable materials. The four main physical properties used to sort foods are size, shape, weight and colour.
6. Drying: Prior to storage or further processing, cereal grains need to be dried. The most cost-effective method is to spread out in the sun to dry. In humid climates it may be necessary to use an artificial dryer. Simple grain dryers can be made from a large rectangular box or tray with a perforated base. The grain is spread over the base of the box and hot air is blown up through a lower chamber by a fan. The fan can be powered by diesel or electricity and the heat supplied by kerosene, electricity, gas or burning biomass. Cereal grains should be dried to 10-15% moisture before storage.
7. Cleaning: Cleaning is the unit operation in which contaminating materials are removed from the edible part of the crops and separated to leave its surface in a suitable condition for further processing. Cleaning is an effective method of reducing food wastage, improving the economics of processing and protecting the consumer. Equipment for cleaning is categorised into wet procedures (for example soaking, spraying, flotation washing and ultrasonic cleaning) and dry procedures (separation by air, magnetism or physical methods). The selection of a cleaning procedure is determined by the nature of the product to be cleaned and by the types of contaminant to be removed. Wet cleaning is more effective than dry methods for removing soil from root crops or dust and pesticide residues from soft fruits or vegetables. Dry cleaning procedures are used for products that are smaller, have greater mechanical strength and possess a lower moisture content (for example grains and nuts). The main groups of equipment used for dry cleaning are, air classifiers, magnetic separators and separators based on screening of food.
8. Grading: This is the assessment of a number of characteristics of a food to obtain an indication of its overall quality. It is normally carried out by trained operators.

9. Decortivating: Decortivating is the process of removing the seeds pod or casing in order to free the seed from any enclosure. This is usually done prior to milling and extraction of oil from oil bearing seeds.

10. Hulling: Hulling is the process of removing the outer husk or shell from grains like rice and barley. This step is essential for making the grains edible and improving their texture. Hulling can be done using hand-operated hullers or mechanical hullers. Mechanical hullers use rotating drums or blades to remove the husk; they are more efficient and can process large quantities of grains quickly. Less rice is broken during hulling if the rice is parboiled first. Rice polishers are available for removing the rice bran after hulling.

11. Pounding/Milling: There are three main types of grain mill: Plate mill; Hammer mill; Roller mill. The choice of mill depends on the raw material and the scale of production. Hammer mills are almost universally used throughout the developing world. Plate mills are widely available in West Africa.

Roller mills are not used at the small scale because of their high cost and maintenance requirements.

The plate mill is usually limited to about 7kW and is derived from the stone mill or quern.

Two chilled iron plates are mounted on a horizontal axis so that one of the plates rotates and the grain is ground between them. The pressure between the two plates governs the fineness of the product and is adjusted by a hand screw. There are manual versions of the plate mill available, though they are arduous and hard work to use.

Small-scale hammer mills range in size from 2kW to 20kW. They consist of a circular chamber in which beaters whirl at a high speed. The milled grain is filtered out through a perforated plate that runs around the edge of the mill chamber. The size of the holes in the perforated plate determines the fineness of grinding of the particles. Most grains can be ground in a hammer mill. Grain for human food is ground to a 1mm particle size while animal food is ground to a 3mm particle size. Hammer mills cannot be used for wet milling.

Roller mills crush the grains rather than milling them into smaller particles. Roller mills are usually used for animal food. It is important to ensure that the grains have the optimum moisture content before milling. If the grain is too dry and hard, it is difficult to break down and requires more energy to convert it into flour. If the grain is too moist, the material sticks to the mill. The optimum moisture content varies between cereal types and with the particular mill being used. Dry grain can be conditioned by soaking in water. Moist grain can be dried before grinding. Different cereal grains have different milling and grinding requirements.

12. Parboiling: Parboiling rice is an optional step, but one that improves the quality of hulling as it results in fewer broken grains. About 50% of all rice grown is parboiled. Parboiling involves soaking and heating the rice which pre-cooks the grains, loosens the hull, sterilises and preserves the rice. At the village level, parboiling is carried out in large pans over an open fire. Rice parboilers, that improve the efficiency of cooking, are available.

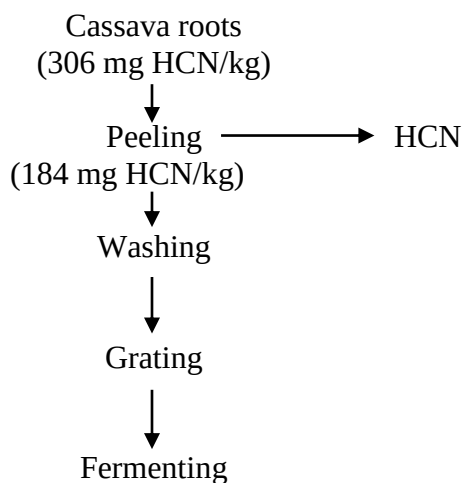
Crop Processing

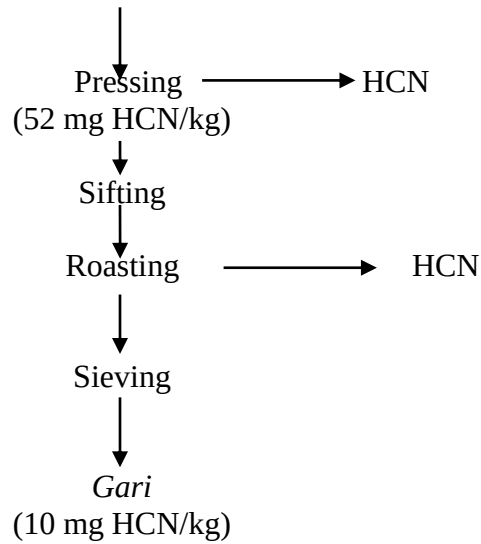
Food processing is a set of techniques used to transform raw materials into finished or semi-finished products or to transform semi-finished products into finished products in order to ensure the availability of the produce all year round and to minimize the postharvest loss of such produce.

Processing of cassava to gari

Fresh cassava roots are peeled and grated. The grated pulp is put in sacks (Jute or polypropylene) and the sacks are placed under heavy stones or pressed with a hydraulic lack between wooden platforms for 3-4 days to express excess liquid from the pulp while it is fermenting. Fermentation imparts an acidic taste to the final product. The dewatered and fermented lumps of pulp are crumbled by hand and most of the fibrous matter is removed. The remaining mass is sieved with traditional sieves (made of woven splinters of cane) or iron or polyethylene mesh. After being sieved, the fine pulp is then roasted in an iron pan or earthen pot over a fire. If the sieved pulp is too wet, it takes longer to roast resulting in a finished lumpy product with dull colour. Palm oil may be added to prevent the pulp from burning during roasting and to give a light yellow colour to the gari. When palm oil is not added, a white gari is produced. Palm oil contains substantial quantities of vitamin A, therefore, yellow gari is 10-30 percent more nutritious and expensive than white *gari*. The garification or conversion rate of fresh roots into *gari* is 15-20%. This value varies with cassava varieties, time of harvesting, age of plant and other environmental factors. Peeling is done mainly by women and children. The peeled roots are grated by women, using a simple traditional grater, but it is done by men if a power driven grater is used. Pressing is done by women in the traditional way but done by men when a hydraulic presser is used. The sieved fermented pulp is roasted almost exclusively by women in a pan or pot on the fire with fuel wood as the energy source.

The conversion of cassava to *gari* illustrates the importance of traditional fermentation. *Gari* produced from cassava is one of the most important staple foods consumed in Nigeria. It is a coarse granular gelatinised flour that is primarily consumed in the form of a meal called eba. Its preparation follows the scheme as illustrated below.





Process flow chart for the production of *Gari*

STEPS INVOLVED IN GARI PRODUCTION

1. Root preparation

The roots are prepared within 48 hours after harvesting. Processing involves removing the ends and chopping the remainder into short pieces of about 15 – 20 cm long. The roots are then peeled by hand.

2. Peeling

The roots are hand peeled to remove the outer brown skin and inner thick cream layer and washed to remove stains and dirt.



Image showing the peeling of cassava roots

3. Washing

Peeled roots are washed with clean water and packed in woven basket, just to allow the water to drain.



Image showing the mass washing of cassava roots



Image showing the washed cassava roots

4. Grating

The peeled roots are fed into a grating machine with revolving blades of 2.5 cm impact cross section. The grating brings the previously compartmentalized cyanogenic glucosides in contact with linamarin, where it is broken down to acetone cyanohydrin and glucose. The mash resulting from grating is dewatered to about 50 % moisture content while retaining at least 70 % of its weight. The dewatering facilitates the subsequent solid state fermentation.



Grating of cassava washed roots



Image showing the surface of a grating drum

5. Fermentation

The fermentation of cassava is the most important step in *gari* production. The cassava mash is inoculated with cassava liquor from a six-day-old fermented mash (backslopping) at a rate of 1 litre of liquor to 45 kg of pulp. This reduces the fermentation time from 6 days to 24 hours. The grated pulp is transferred to a jute sack and left to ferment in solid state. Spontaneous fermentation is also used where the process is initiated without the use of an inoculum. However, due to the lag phase that microorganisms have to go through, fermentation times are longer and the risk of fermentation failure is high. The 3-day old cassava juice used for seeding contains microorganisms in their early stationary phase. Such microorganisms will have synthesised the necessary enzymes to be able to utilise nutrients in the substrate, hence the absence of the lag phase. Thus, through repeated use of inoculums from a previous successful fermentation, the best-adapted strains are selected. The predominant lactic acid bacteria isolated from fermenting cassava products including *gari* are *Lactobacilli*, *Leuconostoc mensesnroides* and *Streptococci*. However, lactic acid bacteria are thought to be responsible for the sensory qualities and texture of *gari*. During the fermentation process, the pH of the mash has been found to decrease from 6.40 to about 4.50 as a result of the production of some organic acids by the action of the microorganisms involved.

6. Pressing/Dewatering

The fermented pulp is placed in a nylon mesh bags and dewatered in a hydraulic press to form a cake of 47-50 % moisture content.



An hydraulic press used in dewatering the fermented pulp

7. Sifting

Sifting involves the separation of the coarse particles i.e. ungrated portion of cassava lumps from the fine and smoother ones. After the moisture content of the mash or pulp has been reduced to about 35 to 40% in the process of dewatering, it is then sieved. After pressing, the dewatered cassava mash is a solid cake, which has to be broken up and sieved to remove the large lumps and fiber and to obtain homogenous product. To know this moisture content, squeeze it in the folds of palm if it is sufficiently dewatered, it will disintegrate. Uniform particle size is important because it makes for a more uniform roasting of individual particles during the frying operation. Indeed, smaller particles takes less time and energy to roast, sieving the final product of *gari* ensures uniformity of the product.



Image showing sifting of *gari*

8. Roasting (garification)

The filtered cake is disintegrated and then roasted in a rotary kiln. When the core temperature in the kiln has reached 260 to 280°C, the cassava mash becomes partially gelatinized with a moisture content of 40%. The gelatinization process requires high heat and low mass transfer. The roasting stage is critical for the swelling of the *gari*.



Image showing the traditional roasting of *gari*

9. Sieving and Packaging

The final product is sieved again to ensure uniformity of the final product and then packaged in a sack or polyethylene bag.



Image showing the final dried *gari*

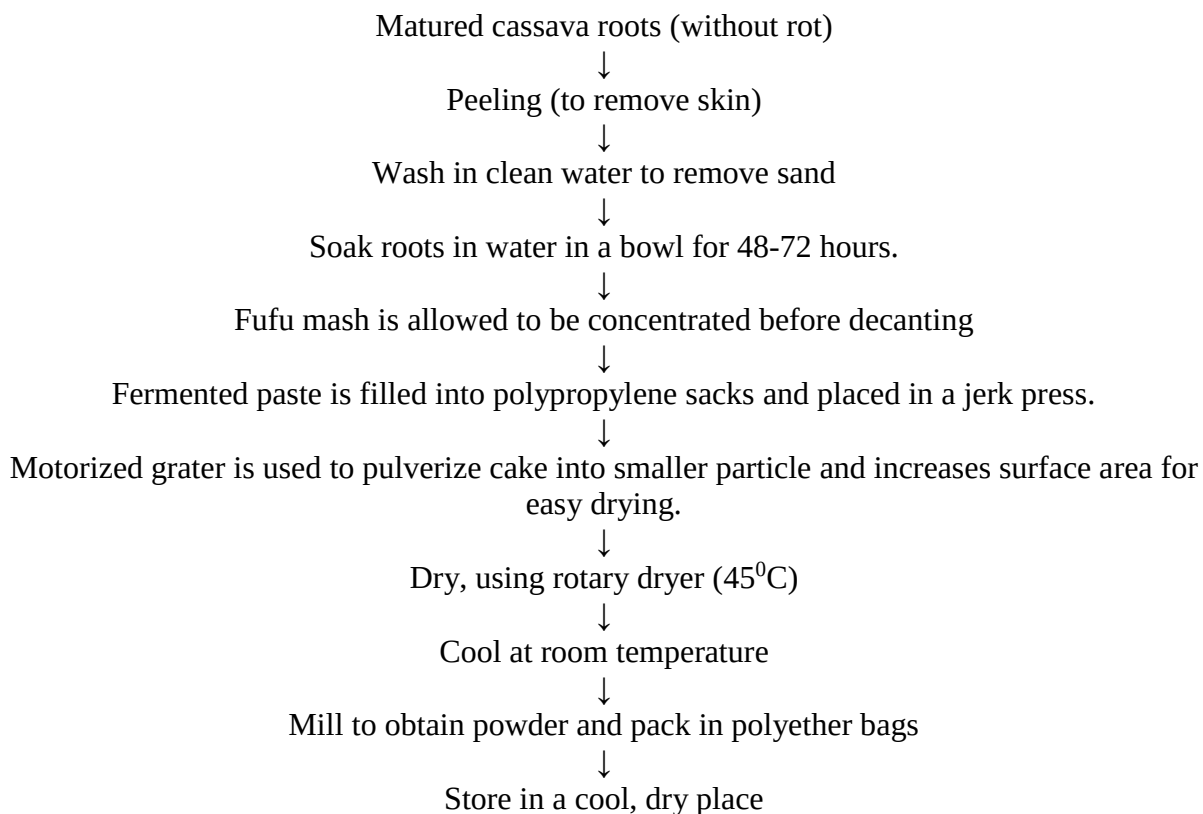
Traditional and Improved Cassava Processing Technologies/Equipment

Processing Stages	Traditional Technology	Improved Technology	Level of Adoption of Upgraded Technology
1. Peeling	Knife made of bamboo, flint or metal	• Mechanical peeler • Motorised peeler • Hand peeler, hand rasper	1%
2. Washing	Local Calabash bowl	Aluminium tank	60-78%
3. Grating	Rough Stone, prickly trunk of palms or sheet / tin iron pierced with nail on one side	Mechanized Grater, Motorised grater, Hammer mill, disk grater, hand grater	89-95%
4. Fermentation	Heavy stone on heavy weighed cloth or nylon bag	Batch fermentation in aluminium locally made hydraulic or mechanical	3-5%
5. Dewatering	Pressing heavy stone on heavy weighed cloth nylon bag (for several days)	Hydraulic jack press, screw press, parallel board press, upgraded traditional press for few minutes.	50-88%
6. Sieving	Woven baskets, suspended cloth pieces holding mash	Improved pulverizer e.g. drum sieve, rotating sieve	2-5%
7. Roasting	Cast Iron pan over wood fire	Upgraded roaster, solar dryer, kiln type dryer	2-5%
8. Sifting	Woven basket	Improved pulverizer and sifter	2-5%

9.Value addition	Frying with palm-oil	Fortification with rich proteneous plants like soybean, amaranth grain, melon seed, okro. Recent development: moringa powder	Soy-gari (1-2%) Amaranth-gari (<1%) Okra-gari (<1%) Moringa-gari (<1%)

Fufu

Fufu is a traditional fermented white paste (porridge) made from cassava. It is ranked next to *gari* as an indigenous food of most Nigerians in the South. *Fufu* is made by steeping whole or cut peeled cassava roots in water to ferment for maximum of three days, during the steeping, fermentation decrease the pH, softens the roots and help to reduce the potentially toxic cyanogenic compound. The process flowchart for the production of *fufu* is shown below. The wet pulp may be molded into balls, 3-5 cm diameter, put in boiling water and stirred thoroughly to obtain a stiff wet pulp of about 0.5-1.0 kg is packed in a plastic or polypropylene bag and marketed in cities in Nigeria, Ghana and Cameroon.

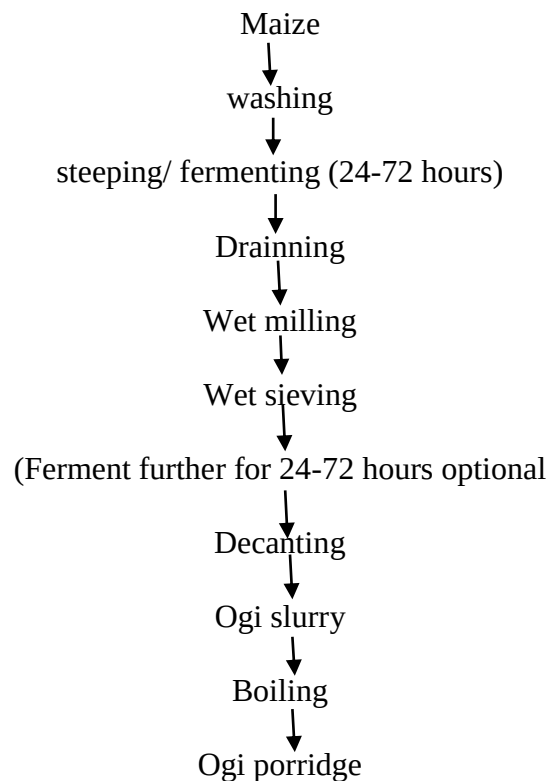


Process flow chart for *fufu* production

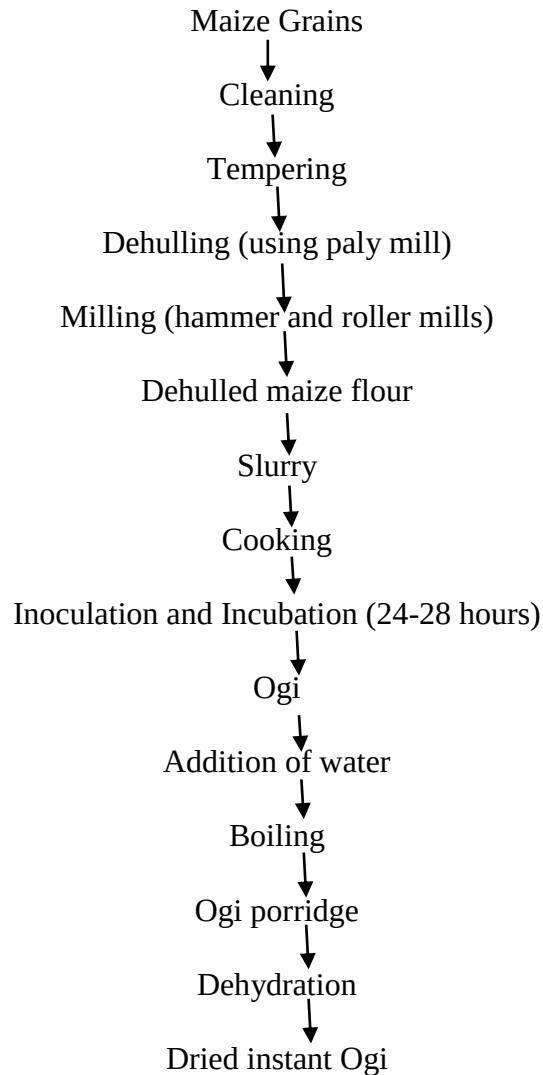
Processing of cereal grains

Ogi

Ogi (acid-fermented cereal gruel) is a staple food of several communities in Nigeria. It is traditionally made from maize, sorghum or millet. Several reports had identified steeping and souring as the two fermentation stages involved in the traditional process of ogi. It is prepared by steeping clean grains in water at room temperature ($25\pm 2^{\circ}\text{C}$) for 48-72 h. The steep water is decanted and the fermented grain is washed with clean water and then wet-milled. The bran is removed by wet sieving and the sieveate is allowed to settle for another 24-48 h, a process referred to as souring during which time fermentation also proceeds and the solid starchy matter, ogi, sediments. The wet ogi usually has a smooth texture, a sour flavor resembling that of yoghurt and a characteristic aroma that differentiate it from starch and flour. The wet ogi can be boiled at 8-10% total solids into a porridge or pap which serves as weaning food for infants, breakfast for children and convenient meal for the convalescence. It is usually marketed as a wet cake wrapped in leaves and polyethylene nylon is also in use recently. The wet cake can also be cooked to a stiff gel or cold pap called "agidi" or "eko" which commonly package in a leaf before consumption. This same product is often eaten along with meat, stew, vegetable soup, steamed bean cake ('moin-moin') or fried bean cake (Akara). The physical and biochemical qualities of ogi are influenced by the type of cereal grain, fermentation or souring periods and the milling method.



Traditional Nigeria Ogi manufacture



Improved Nigeria Ogi manufacture.

Other consumable products processed from maize include tuwo/inioka, donkunnu, maasa and wainna, cous cous, akple, gwater, nakia, dambo alubosa, abari/sapala, egbo, donkwa, popcorn, ajepasi, aadun, kokoro, cooked or boiled maize and roasted maize

Processing and Utilization of Cowpea

Cowpea (*Vigna unguiculata*) is a common name for large seeds of the general *Fabaceae* (formerly *leguminosae*). Cowpea is of considerable importance in Nigeria and in many Africa countries as a nutritious leguminous crop providing an alternative source to animal protein. Cowpeas are frequently consumed in West Africa as fried "Akara balls and steamed moin-moin", both of which are prepared from ground beans.

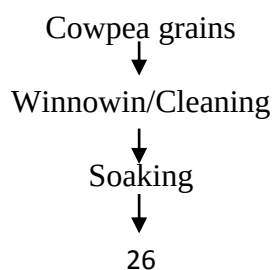
Akara (Bean cake/ball)

In Western Africa, a popular way to eat cowpea is to make a batter from which fritters are manufactured. These fritters (known as accra, akara, akla, binch akara, bean balls, kosai, koosé, and kwasi) are commonly prepared at home for breakfast, for snacks, or as an appetizer or side dish. They are also fast food, sold by vendors on the street, in marketplaces, and at bus stations. The procedure involves cleaning the black-eyed peas in running water. Soak them in water for at least a few hours or overnight. After soaking them, rub them together between the hands to remove the skins. Rinse to wash away the skins and any other debris. Drain them in a colander; crush, grind, or mash the black-eyed peas into a thick paste. Add enough water to form a smooth, thick paste of a batter that will cling to a spoon. Add all other ingredients (except oil). Some people allow the batter to stand for a few hours (overnight in the refrigerator); doing so improves the flavour. Heat oil in a deep skillet. Beat the batter with a wire whisk or wooden spoon for a few minutes. Make fritters by scooping up a spoon full of batter and using another spoon to quickly push it into the hot oil. Deep fry the fritters until they are golden brown. Turn them frequently while frying; Serve with an "african hot sauce" or salt, as a snack, an appetizer, or a side dish.

Moyin-Moyin (Steamed bean pudding)

"Moyin-Moyin" (also called Moin-Moin, Moi-Moi), a sort of savoury bean pudding, is a unique and delicious way to prepare black-eyed peas or other beans. The traditional way to cook "Moyin-Moyin" is to wrap it in leaves (such as banana leaves) and steam it. In modern Africa it is often cooked in empty tin cans, but it can also be made in muffin pans (muffin tins). There are many variations of "Moyin-Moyin".

To prepare moin-moin, clean the black-eyed peas in water in a large pot. Soak them in water for at least an hour or overnight. After soaking them, rub them together between the hands to remove the skins. Rinse to wash away the skins and any other debris. Drain them in a colander. If the beans have soaked only a short time, they may be cooked in water over a low heat until they are partially tender. Crush, grind, or mash the black-eyed peas into a thick paste. Slowly add enough water to form a smooth, thick paste. Beat with a wire whisk or wooden spoon for a few minutes. A tablespoon of oil may be added. In a separate container combine all other ingredients and crush and stir them together until they are thoroughly mixed. Add the other ingredients to the black-eyed pea paste and stir to make a smooth mixture. Grease the muffin pans (or tin cans). Scoop the "Moyin-Moyin" mixture into your pans (or cans), allowing some room for it to rise while cooking. Place the pans (or cans) in a baking dish partially filled with water. Bake in a medium-hot oven for about a half-hour. "Moyin- Moyin" in tin cans can also be steamed in a large covered pot on a stove. Check for doneness with a toothpick or sharp knife, as one would for a cake; An alternate cooking method is the following: Wrap the Moyin-Moyin mixture in banana leaves or aluminium foil to make small packets. Cook the packets by steaming them in a large pot, using a rack to keep them out of the water. It may be served hot or at room temperature.



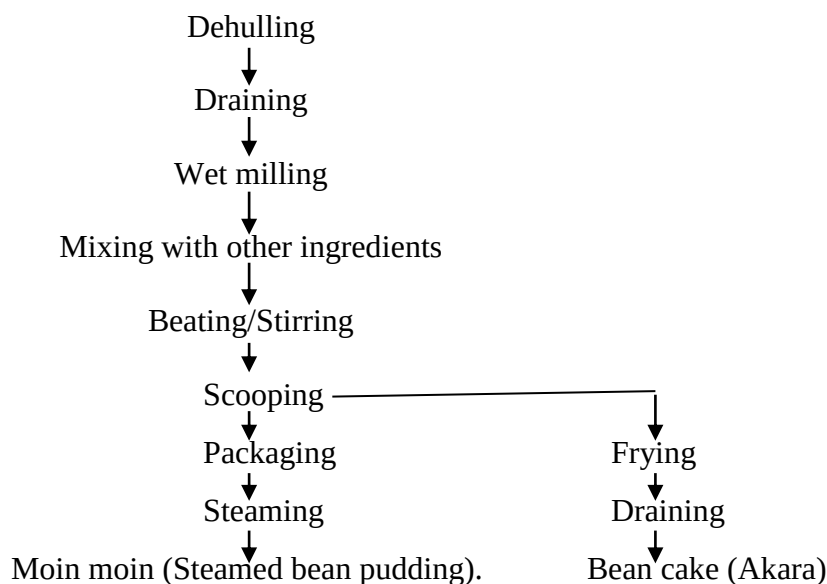


Fig 3.3. Flowchart for *moin moin* and *akara* production from cowpea grains.

AGRICULTURAL PROCESSING MACHINES

1. Shelling Machines: Shelling machines are machines which apply mechanical force in removal of kernel from its casing (or enclosure) in order to free it from any attachment or confinement e.g. removal of palm kernel from its nuts, removal of melon and moringa kernels from their shells. The principle of operation of shelling machines depends on the nature of the agricultural materials. For instance most motorized shellers use impact force in removing kernels from their encasement. This involves the application of mechanism such as flat bar or rasps bar cylinder with concave screen. In this case the seeds are loaded into the hopper from where it moves into the shelling chamber. It is then beaten by the rotating cylinder to free the kernels from their casing. During this process, the kernels are being separated from their host and pass through the screen to be discharged at the chute. It should be noted that the principle of shelling depends relatively on the engineering properties of biomaterials to be shelled, as it varies from one type to another. Other shelling machines include melon sheller, palm nut cracker, bush mango seed shelling machine etc.

2. Thresher: This uses the same principle as shelling machine to operate. Threshing means detachment of corn from their panicle. Threshing is accomplished by impact action in most conventional threshers. A rotating cylinder (spike-tooth cylinder) and a concave screen are utilized to accomplish threshing. As the cylinder rotate, the seed is detached by impact force from its panicle and drops on a reciprocating screen which separates the straws from the seed. Types of threshers include wheat, sorghum, millet guinea corn threshers and so on. The components of most threshers comprises of the following:

a. Hopper: to hold the agricultural materials before threshing b. Spike-tooth cylinder: to detach the seed from its panicle. c. Concave/Screen: to separate seeds from undesirable materials d. Chute: to discharge processed agricultural materials.

3. De-hullers: A de-huller is a machine used in removing outermost thin-layer covering of a seed. De-hulling is a very important unit operation in food processing. De-hulling is the removal of the outer pericarp and testa (hull) during processing of cereal grains, grain legumes, nuts and oilseeds. Besides nutritional and aesthetic enhancement, de-hulling reduces the bulk density of the plant material, thereby facilitating better temperature management and control of other rate transfer processes in downstream processing. This unit operation is necessary in order to have products with desirable qualities. In the preparation of steamed bean cake, de-hulling of cowpea is very important in order to improve its appearance. Most de-hullers use frictional force in removing the outer thin layer of seed. This could be done using a serrated surface cylinder or grinding discs with a concave screen. The serrated cylinder rubs against the crop to remove its outer cover after which it is reduced to powdery form and passes through the screen to be removed while the de-hulled seed is discharged through the chute. The components of a de-huller are:

a. Hopper- to hold the agricultural materials b. Serrated Cylinder or Grinding Disc- to de-hull the seeds c. Concave/Screen: for separation of grain from hulls. d. Chute: to discharge the processed agricultural materials

4. Decorticator: A decorticator is a machine that is used in the removal of outer coat of any agricultural materials usually pods. The process of decortication is synonymous to shelling. The word decorticate comes from the Latin verb decorticare derived by adding the prefix de (meaning to be rid off) to the noun Cortex. The cortex refers to the outer layer of a botanical or anatomical object, often specifically to a layer which lies just beneath the very outer surface, and to general features such as bark, husk, shell or rind. Conventionally, decortication encompasses dehulling, husking, cracking, splitting, popping and shelling. Decortication process is more generic than shelling process. In this case a decorticator could be used to remove the bark of a stem, seeds pod etc. A decorticating machine is a very versatile machine. Most decorticators apply shear and impact force to bring about decortications. This process is one of the most delicate operations in crop processing because the structure of the seed must not be tampered with or deformed. Types of decorticators include cowpea, groundnut and jatropha fruit decorticators.

STORAGE OPTIONS FOR ANNUAL CROPS AND PLANTING MATERIALS

Storage options can be broadly classified under the Traditional and Modern structure options.

Traditional storage structures: They are used for short term (6 months) and small-scale (1 ton) storage and sometimes medium term (12-18 months) and medium scale (2-50 tons) storage. They are mostly made of unrefined local materials. They include:

1. Barn for storing fresh yams
2. Cribs for storing grains
3. Bins
4. Bag/ Polythene bag
5. Rhombus
6. Shelf
7. Pit/ Underground storage

8. Plastic containers
9. Guard
10. Earthen pot

Rhombus and traditional cribs: These are used for storing grains such as guinea corn and millet that have not been threshed. Rhombus is cylindrical in shape while a crib has a rectangular shape. They are made of palm frond leaf, clay, tree stem and bamboo. Sometimes heat is introduced at the lower base to ensure drying. The main disadvantage of these storage options is moisture build up as a result of rain and micro-organism infestation.

Barn, Shelf and Pit: These are mostly used for storing root and tuber crops such as cassava, yam and cocoyam. Barn and shelf are suitable for onion and carrot.

Barn is used for storing fresh yams. It is made with low walls that have openings for ventilation. It should be rodent -proof to prevent rodent infestation. Yams are either tied on poles in heaps or raised platforms. Other crops like maize and guinea corn can be stored in a barn.

Plastic containers, Guard, Polythene bag and Earthen pot: These structures are used at household and peasant levels for the storage of grain. The stored items are locally preserved with wood or bone ash or powdered pepper.

Bags are used for storing grains like beans, rice e.t.c. They are made airtight, pest-free and kept in a clean store. The bags should be neatly stacked on wooden racks away from the walls and off the floor. The major limitation of jute bag storage is that the bag does not provide protection against rat or insect attack.

Drums/ Bins: They are used to store small grains in an airtight container and sealed up. This kills insects that may be present in the grain. Sometimes, chemicals are added to preserve the stored grains. The drum should be kept out of the sun, otherwise, the hot sun hitting the metal drum will make the grain very hot, sweat and respire faster and this can cause spoilage.

Modern storage options: They are mostly with large capacity and for long term storage and there is better regulation of the storage environment. The structures are made of improved and refined materials. They include:

1. Warehouse: This is used for storing bags or bulk products like grains and flour.
2. Improved crib: This is an improvement over the traditional crib in terms of design, capacity and construction material.
3. Silos for storing grains: This is a cylindrically shaped structure used for bulk storage of shelled grains in large scale and for long term.
4. Canning: This is the process in which food is sealed in air-tight containers. It helps in keeping away bacteria, unwanted enzymes and oxygen that can destroy the food. Preservatives are sometimes added to the food.

5. Cold storage e.g refrigerator: This used to store highly perishable crops like fruits and vegetables in cold chamber. Cold temperatures inactivate enzymes and reduce activities of spoilage organisms like bacteria and fungi.

PASTURE /FORAGE AGRONOMY

Pasture: This is any area of land containing herbage which could be grasses or legumes needed for feeding farm animals. They may allow the animals to graze directly on the herbage or the herbage can be harvested and fed to the animals in the stall. Where the herbage is consumed fresh it is called forage crop, while if harvested and preserved in fresh form as silage and dried form as hay is called fodder crop.

Definition of pasture based on purpose

Pasture can be defined in many ways depending on purpose:

- Pasture is defined as an area of land occupied with forage plant species either natural or planted by man for the purpose of providing qualitative feed to livestock at the right quantity.
- It could refer to any forage species valuable as livestock feed
- It can also refer to an act of feeding; grazing or browsing animals on pasture plants or pastures.

Forage and Forage Crops

Forages are plant materials eaten by grazing livestock. The term forage, historically, meant only plants eaten by the animals directly as pasture, crop residue, or immature cereal crops, but it is also used more loosely to include similar plants cut for fodder and carried to the animals, especially as hay or silage.

Classification of Pasture Plants

Characteristics of grasses and legumes as pasture species

(1) Grasses

Grass belongs to the broad family Poaceae or Gramineae. There are over 10,000 species of grasses, these are grouped into 650 genera. Of this 650, only 12-15 of tropical grasses that are widely used in sown pastures. Many grasses have similar features, although, growth forms can vary. These are

- i. Tufts/Tussocks/Bunches
- ii. Prostrate/Creeping/Straggling.

Most of these creeping types have stolon or rhizome. Those with stolon creep along the ground while those that creep below the ground have rhizome.

Some of the important characteristics of Grasses are

1. Generally, tropical grasses have erect culm.
2. Most grasses are propagated from seed or split from existing parts.
3. Stolon and rhizomes tend to form sward rather than tuft.
4. There are three main types of grasses used as forage.

- i. **Bunched or tufted grasses:** These form a clump or tuft. If allowed to set seeds fall to the ground and germinate e.g Guinea grass (*Panicum maximum*)

- ii. **Stoloniferous grasses:** These send out stems or stolon along soil surface rooting at the nodes and producing new shoots e.g. para grass, *Brachiaria mutica*
- iii. **Rhizomatous grasses:** These send out stems or rhizomes below the soil surface. E.g. Quantamala grass (*Tripsacum laxum*).

Tufted or Bunched Grasses

- 1. Established from seeds or root stalk
- 2. Spread slowly
- 3. Should not be over grazed
- 4. May be deeply rooted and drought resistant
- 5. They have variable productivity
- 6. Combine well with climbing legumes
- 7. Growth is usually moderate to tall

Stoloniferous grasses

- 1. Established from cuttings
- 2. Spread quickly to cover ground.
- 3. Usually tolerant of heavy grazing
- 4. Some are deeply rooted and drought resistant
- 5. Generally productive
- 6. Difficult to maintain a good legume percentage with the aggressive species
- 7. Height is usually short.

(2) Legumes

Legumes belong to the broad family Fabaceae (Leguminosae). They are divided into three distinct group or sub families

- 1. **Mimosoideae:** This group contains several browse plants and woody species.
- 2. **Caesalpinioideae:** These contain about 10,800 species, most of which are trees of tropical savanna of forest Africa and Asia.
- 3. **Papilionoideae:** Almost all the important pasture legumes belong to this group. It includes up to 200 genera and 12,000 species distributed throughout the world. Like the grasses only small number of legume is used commercially in pasture production.

Characteristics of legumes are

- 1. Some are creeping and root at the node and stolon. They can reach sunlight by climbing on the associated grasses e.g. centro, calopo etc
- 2. On the root systems of most legumes, nodules may develop in which nitrogen fixing bacteria of the genus rhizobium form a symbiotic relationship with the plants (the legume nodule forms symbiotic relationship with the plants and fix-nitrogen through the root nodules)
- 3. Large amount of nitrogen (N) may be fixed through root nodules.
- 4. They have high protein content

Important characteristics that form choice of the species

These include

- 1. Yield
- 2. Palatability

3. Nutritive value
4. Propagation
5. Rapidity of establishment
6. Ability to compete with weed, pest and diseases
7. Ability to associate with other pastures species
8. Adaptability to local soil and climatic conditions
9. Management levels
10. Shad tolerance etc.

Pasture plants can be classified into the following:

- a) **Grazed Plants:** These are short herbs that are consumed with both leaves and stems together. Examples include: *Brachiaria ruziziensis*, *Sorghum alnum* (Columbus grass), *Arachis pintoi* and *Digitaria smutsii*.
- b) **Browed Plants:** These are woody plants whose leaves and twigs (soft stems and buds) are eaten while the woody stem is left uneaten. Examples include: *Leucaena leucocephala*, *Gliricidia sepium* and *Gmelina aborea*.
- c) **Annuals:** These are plants that complete their life cycle within one growing season and survive the dry season as seeds. Example include: *Chloris gayana*, *Brachiaria mulato*, *Pennisetum purpureum* and *Hypherrania rufa*.
- d) **Biennials:** These are plants that complete their life cycle in two seasons. First season is for vegetative growth and second season for reproductive growth. Example include: *Panicum maximum* (Guinea grass), *Andropogon tectorum* (Southern Gamba grass) and *Cynodon dactylon* (Bermuda grass).
- e. Pasture Weeds: When a pastureland is left unattended in terms of management practices for years, there is a tendency for weeds to invade the pasture. These weeds could be controlled manually by removing them with hand or through cutting with cutlasses. The farmer may decide to use chemicals such as pre or post- emergence herbicides at the time of pasture establishment, mainly for broad-leaved weeds. Weeds reduce the quality of pasture especially when they invade large portion of land in the pasture. They also compete vigorously with the pasture plants for space, nutrients, light intensity and moisture content. Examples of common pasture weeds in Nigeria include:
Amaranthus caudatus (waterleaf), *Crinum ornatum* (Lily), *Solanum incanum* (Graden egg), *Cyperus rotundus*; Kyllinga sp. (Nut Grass), *Cucurbita maxima*, *C. pepo* (Pumpkin) etc.

Major grasses

Guinea grass

- Family Name: Poaceae
- Botanical Name: *Panicum maximum*
- Genus: There are up to 500 species distributed throughout the warm tropics of the world. Warm tropics with high temperature and rainfall that allow the plant to establish properly.
- Origin: The plant is native to tropical Africa.
- Distribution: It is distributed throughout the warm tropics and it is one of the most important grazing and fodder grass.
- Form
 1. They are erect, vigorous in growth, tufted perennial stem which growth varies from 0.5 – 4.5m

2. They exercise great variability in species and growth habit but the genus *Panicum maximum*

Adaptation: They are drought resistance and perform well in areas with high Temperature and rainfall above 1300mm per year. Land this makes it to grow and survive during dry season)

- Propagation
Propagation is by means of seeds or part of the Vegetative root stocks.
- Grazing value:
They are fairly tolerant of heavy grazing but should not be grazed below 15-20cm

Yields

They are productive, but when they are shaded, especially when planted with tree crops there will be reduced yield except the spacing regim is managed properly.

Mixtures/Combination: They mixes or combine well with *Centrosema pubescens*, *Calopogonium mucunoides*, *Leucaena leucocephala*, etc.

Napia fodder or elephant grass

Family Name: Poaceae

Botanical Name: *Pennisetum purpureum*

Origin:- Native to tropical Africa

Distribution: They grow throughout tropical Africa.

Form

- It is a robust perennial
- They have erect (cum), and form large broad crumbs at the base and more difficult to control when not used as pasture
- They spread at the stem base
- They root at the node or by short rhizome
- The plant is usually very tall, ranging from 2-6m
- They have many varieties which vary in height, size of the leaves, thickness of stem and the flowering behavior.
- The leaves are usually long from 50 – 100cm and narrow with prominent midrib and rough edges.

Adaptation: They are suited to areas with rainfall of about 1000mm, drought tolerant but cannot tolerate flooding or water logged soils.

Propagation: This is by means of divided stem cuttings or root stocks. They are generally propagated through vegetative means because their seeds are not viable.

Mixtures/Combination

They can mix or combine well when grown with vigorous climbing legumes e.g *Peuraria phaseoloides*, *centrosema pubescens*, or in alternating role with *leucaena leucocephala*.

Axonopus spp

There are two species commonly used

1. *Axonopus affinus* (Narrow leaf carpet grass)
2. *Axonopus compressus* (Broad leaf carpet grass)

Botany

- *Axonopus spp*, is **creeping perennial grass, spreading by stolons and forming dense low growth.**
Origin: Native or originated from Central and South America tropics.
- Ecology: They can thrive well on a range of soil conditions
- Botany: They are shallow rooted and may die back in dry season.
Yield: Low dry matter yield of about 2-2.5t/ha

Grazing value

- They can withstand heavy grazing pressure and trampling.
- Livestock feeding on them produce moderate live weight gain.

Major legumes**Centro**

(1) *Centrosema pubescens*

Family: Fabaceae

It is the most widely distributed legume and widely used as pasture legume. It is **found growing freely in ranches.**

Origin: Native to central and South America

Form:

1. Growth is Vigorous
2. Form balanced mixture with grasses
3. They are trailing with smooth fine stem
4. They are climbing perennial herbs
5. Trifoliate with slightly hairy leaf
6. They have distinctive white purple flower (cleistogamous)
7. They form linear pods 7.5 – 15cm with red broom seeds, streak with black

Ecology

- Well adapted to wet tropical Conditions with rainfall in excess of 1750mm but can grow in Areas with rainfall as low as 750 – 1000mm
- It is fairly drought tolerant due to deep roots.
- It combines well with both tufted and creeping grasses
- It is moderately palatable
- Withstands fairly heavy grazing
- They are hard seeded, so they need seed treatment by scarification
- Propagated from seeds.

Centrosema pascuorum

Family – Fabaceae

Origin – Native to central and South America

Form

- They are vigorous
- They are trailing herbaceous legume
- They are short lived, under high temperature and low rainfall they die off
- Fairly tolerant to drought and Adapted to acid soil

- Combine well with tall grasses
- They produce high seeds

***Stylosanthes hamata* (Caribbean stylo)**

- Family – Fabaceae
 - Common name: Stylo
- They are short lived perennials and produce a lot of seeds.

Origin: Native to West Indies

Form

- Produces high seed
- Rich in protein
- Readily propagated from seed
- They nodulate readily with active rhizobium of the cowpea type
- Drought resistance
- Maintain a high nutritive value during the dry season

Ecology

- Grows on wide range of soils and
- Tolerate high acidity
- It is moderately resistance to (Anthracnose) *Colletotrichum truncalum*.

Morphological characteristics of grasses, legumes and shrubs

Pasture and forage crops have unique characteristics that can easily be used to identify them in the field.

Morphology of Grasses

Grasses have two different parts. These include:

- The root
- The shoot

The root of grasses is fibrous in structure and it occurs in tussock or bunch depending on the specie.

Morphology of Legumes

Legumes are divided into four different parts:

- Root
- Branches
- Leaves
- Flowers

Morphology of Shrubs

Shrubs have the same morphological structure with forage legumes. The only difference between them is that shrubs contain more woody stems than forage legumes and they have more leaves, flowers and seeds. Also, the height of shrubs could be up to 4 meters, unlike forage legumes whose height might be 1 meter or even less.

Types of Pastures

There are two main types of pasture

- 1. Natural Pastures:** These are pastures that are not planted but found growing wild or naturally e.g. Natural grass Land. Where animals can graze pasture species freely.
- b. **Artificial or Improved Pastures:** These are pastures that are deliberately established by man for feeding livestock. Improved pastures are pastures that have undergone breeding, selection and improvement for better yield and livestock performance.
- c. **Permanent Pastures:** These are pastures that are permanently maintained for feeding animals. They contain perennial species which could be grasses and legumes that are resistance to trampling.
- d. **Semi- permanent pastures:** These are pastures that are established for a few years to cover soil and provide feed for livestock. Such pastures are converted into cropping lands after a few years.
- e. **Ley or Rotation Pastures:** These are pastures that are established and grazed by animals for two or three years after which the land is reverts to cropping.
- f. **Annual or Temporary Pasture:** These are pastures that are established and harvested within one year.

Importance of pastures

1. Can be used as food by Man: e.g. wild growing pods from legumes and succulent parts of grasses have been used by man for food.
2. They help in the conservation of soil through aggregate stability, which is achieved through binding together of soil by roots.
3. The help in Water Conservation by catching precipitate in the form of rain
4. They add to the diversity and beautification of Agricultural landscape.
5. Improve Soil Structure through aggregation of soil particles which in turn influences soil water regime and soil aeration.
6. Forages have protective action through the reduction of tremendous energy of rain drop size thereby reducing run off and soil erosion.
7. They Improve Soil Fertility through fixation of nitrogen from the atmosphere and from the dead decaying materials add Organic matter which in turn add nutrient to the soil.
8. Aid pollution control by improving infiltration rate in the soil and reduction of runoff. Grass crop is associated with reduction in the delivery of sediments and other pollutant to stream irrigation channels.
9. Animal by Product by planting grasses on waste land or marginal land we are converting such land into animal product. Marginal lands are soils that cannot be used to produce crop in terms of fertility but can be used to raise forages for animals.

Factors to be considered for pasture establishment

Pasture should be established with some consideration, for it to be successful. Some of the factors to be considered before establishing pasture are: Land, plant species, high nutritive value, fast growth, ease of establishment (**Persistency, Finance and Environment**)

Procedure for pasture establishment

The procedure follows the same method as that of arable crop farming except that forage farming requires more skillful practice. The procedure includes the following: Land and soil survey, land preparation, soil treatment, planting operation, seed processing, Concentration of pasture seeds, seed treatment

Forage conservation techniques

Forage conservation is one of the methods of pasture management in Nigeria. Excess forage is conserved during the period of abundance to be used during period of scarcity. In order to avoid excessive spending on purchase of expensive concentrate feeds, farmers harvest forages during peak period of growth and preserve in form of hay, silage or stacks.

Hay: This is harvested forage crop which has recognized feed value, not coarse, woody and contain, not, more than 25% moisture, preserved by drying (curing) and used primarily for feeding ruminant.

Silage: This is a preservation of feed which refers to fermentation of forage crop used for feeding animals either throughout the year or during the period of feed scarcity. It is one way of tiling forage animals.

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